

$^{32}\text{S}(^3\text{He}, n\gamma)$ 1974Gr19, 1974Be18, 1972Ca22

1974Be18: an 8-MeV ^3He beam was provided from the ANU EN tandem accelerator and impinged on cooled sulphur targets.

Neutrons were detected using an NE213 liquid scintillator and γ rays were detected using a Ge(Li) detector and a NaI(Tl) detector. Measured E_γ , I_γ , $n\gamma$ -coin, $n\gamma(\theta)$. Deduced levels, J, γ -ray branching ratios; deduced lifetimes for 4 levels using the oppler-Shift Attenuation Method (DSAM).

1974Gr19: 8-12-MeV ^3He beams were provided from the 7-MV Van de Graaff accelerator at the University of Freiburg and impinged on natural ZnS targets. Neutrons were detected using an annular NE213 liquid scintillator and γ rays were detected using a Ge(Li) detector and 5 NaI(Tl) detectors. Measured E_γ , I_γ , $n\gamma$ -coin, and $n\gamma\gamma$ -coin. Deduced 10 levels, J, γ -ray branching ratios, and mixing ratios; deduced lifetimes for 5 levels using DSAM.

1972Ca22: an 8.5-MeV ^3He beam was provided from the Oxford University tandem Van de Graaff generator and impinged on Ag_2S targets. Neutrons were detected using a NE102 liquid scintillator and γ rays were detected using a Ge(Li) detector. Measured E_γ , I_γ , E_n , I_n , and $n\gamma(\theta)$. Deduced lifetimes for 3 levels using DSAM.

 ^{34}Ar Levels

$E(\text{level})^\dagger$	J^π	$T_{1/2}$	Comments
0.0	0^+		
2091.0 4	2^+	139 fs 35	J^π : from the Adopted Levels. Spin=2 confirmed by $n\gamma(\theta)$ in 1974Gr19. $T_{1/2}$: from lifetime $\tau=200$ fs 50; weighted average of 330 fs 80 (1974Be18, DSAM), 200 fs 60 (1974Gr19, DSAM), and 150 fs 50 (1972Ca22, DSAM).
3287.5 6	2^+	104 fs 42	J^π : from the Adopted Levels. Spin=2 confirmed by $n\gamma(\theta)$ in 1974Gr19. $T_{1/2}$: from lifetime $\tau=150$ fs 60; weighted average of 215 fs 70 (1974Be18, DSAM), 130 fs 60 (1974Gr19, DSAM), and 120 fs 65 (1972Ca22, DSAM).
3873 3	0^+	>416 fs	J^π : from the Adopted Levels. Spin=0 confirmed by $n\gamma(\theta)$ in 1974Gr19. $T_{1/2}$: from lifetime $\tau>600$ fs (1974Gr19, DSAM).
4127.4 11	$(2)^+$	<243 fs	J^π : spin=2 from $n\gamma(\theta)$, but did not exclude spin=3 or 4 (1974Gr19); spin=4 ruled out by 840γ , M1+E2, to 2^+ . $T_{1/2}$: from lifetime $\tau<350$ fs (1974Be18, DSAM). Other: $\tau>400$ fs (1974Gr19, DSAM).
4513.5 8	3	201 fs 55	J^π : spin=3 from $n\gamma(\theta)$ (1974Gr19). $T_{1/2}$: from lifetime $\tau=290$ fs 80; weighted average of 390 fs 180 (1974Be18, DSAM), 260 fs 80 (1974Gr19, DSAM), and 300 fs 80 (1972Ca22, DSAM).
4631 ‡ 4			
4865 ‡ 4			
4967 ‡ 4			
5255 ‡ 4			
5542 ‡ 4			

† For $E(\text{level})$ below 4.6 MeV, from a least-squares fit to γ -ray energies.

‡ From 1974Gr19 based on γ data without reporting γ energies.

 $\gamma(^{34}\text{Ar})$

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.	δ	Comments
2091.0	2^+	2090.9 4	100	0.0	0^+			E_γ : weighted average of 2091.1 3 (1974Be18) and 2090.0 7 (1972Ca22). $A_2=+0.48$ 9 (1974Be18).
3287.5	2^+	1196.5 4	89 5	2091.0	2^+	M1+E2	+0.12 5	E_γ : weighted average of 1196.6 4 (1974Be18) and 1195.8 14 (1972Ca22). I_γ : weighted average of 92 5 (1974Be18), 90 7 (1972Ca22), and 85 5 (1974Gr19). Mult., δ : D+Q from $n\gamma(\theta)$ (1974Gr19); M2 ruled out by RUL. $A_2=+0.56$ 10 (1974Be18). E_γ : from 1974Be18.
		3286 4	11 5	0.0	0^+			

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$^{32}\text{S}(^3\text{He}, n\gamma)$ **1974Gr19, 1974Be18, 1972Ca22 (continued)** $\gamma(^{34}\text{Ar})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.	δ	Comments
								I_γ : weighted average of 8 5 (1974Be18), 10 7 (1972Ca22), and 15 5 (1974Gr19).
3873	0^+	585	<30	3287.5	2^+			I_γ : from 1974Be18.
		1782 3	>70	2091.0	2^+			E_γ, I_γ : from 1974Be18; other: $I_\gamma=100$ (1974Gr19).
4127.4	$(2)^+$	839.9 9	90 5	3287.5	2^+	M1+E2	+0.52 15	E_γ : weighted average of 840.1 9 (1974Be18) and 839.1 18 (1972Ca22).
								I_γ : others: >50 (1974Be18), 100 (1972Ca22).
								Mult., δ : from $n\gamma(\theta)$ in 1974Gr19; E1+M2, E2+M3, M2+E3 ruled out by RUL.
		2037	10 5	2091.0	2^+			I_γ : other: <50 (1974Be18).
		4128	<10	0.0	0^+			
4513.5	3	1226.0 6	90 10	3287.5	2^+			E_γ : weighted average of 1225.5 6 (1974Be18) and 1226.7 7 (1972Ca22).
								I_γ : others: >90 (1974Be18), 100 (1972Ca22).
								$A_2=-0.38$ 24 (1974Be18).
		2422	10 3	2091.0	2^+			I_γ : others: <10 (1974Be18), <20 (1972Ca22).
4631		2540		2091.0	2^+			I_γ : >50 (1974Gr19).
4865		2774		2091.0	2^+			I_γ : >50 (1974Gr19).
4967		841		4127.4	$(2)^+$			I_γ : <20 (1974Gr19).
		2876		2091.0	2^+			I_γ : >50 (1974Gr19).
5255		3164		2091.0	2^+			I_γ : >50 (1974Gr19).
5542		911	35 10	4631				
		1029	65 10	4513.5	3			

[†] E_γ values without uncertainties are from level-energy differences.[‡] From 1974Gr19, unless otherwise noted.

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Level Scheme

Intensities: % photon branching from each level

